

iClusterBayes Summary Report

Automated Report

Hyperparameters

```
cat('sdev = 0.025, beta.var.scale = 0.1\n')
```

```
## sdev = 0.025, beta.var.scale = 0.1
```

Key Summary Statistics

```
cat(paste0('```\n', '

--- Summary for K = 1 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.4089  0.4350  0.4428  0.4425  0.4511  0.4794

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.09111 0.13444 0.13070 0.16667 0.89667
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.09333 0.13778 0.13323 0.16944 0.79833
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005000 0.008333 0.102017 0.031111 1.000000
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Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0005556 0.0050000 0.0083333 0.0775255 0.0240278 1.0000000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0061111 0.0150000 0.162485 0.118889 1.000000

--- Summary for K = 2 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.1583  0.1839  0.1911  0.1914  0.1983  0.2228

beta.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.04778 0.09611 0.09797 0.13444 0.90111
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1

gamma.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.0500  0.1000  0.1006  0.1389  0.6800
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.001667 0.057778 0.414611 1.000000 1.000000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0005556 0.0050000 0.2488552 0.3463889 1.0000000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.1233  1.0000  0.6974  1.0000  1.0000

--- Summary for K = 3 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.04833 0.07111 0.07722 0.07806 0.08444 0.13111

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beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.03111 0.07389 0.07827 0.11167 0.90222
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.03333 0.07778 0.08166 0.11667 0.61222
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.5650  0.7033  0.6794  1.0000  1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.000000 0.002222 0.255317 0.431944 1.000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.4156  1.0000  0.7528  1.0000  1.0000

--- Summary for K = 4 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.01722 0.02944 0.03444 0.03493 0.03833 0.09722

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.02333 0.06111 0.06668 0.09444 0.76667
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :

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      Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
      1         1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.00000 0.02500 0.06556 0.07076 0.10111 0.61278
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000 0.7044 0.9494 0.7583 1.0000 1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
      1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000 0.0000 0.0000 0.2793 0.8004 1.0000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000 1.0000 1.0000 0.8672 1.0000 1.0000

--- Summary for K = 5 ---
Z.ar:
      Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.008889 0.024444 0.030000 0.031443 0.037222 0.105556

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.009444 0.038333 0.049916 0.076111 0.830556
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
      1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
      1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
      1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
      1         1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.00000 0.01000 0.04222 0.05349 0.08222 0.55333
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000 0.3744 0.7767 0.6515 1.0000 1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
      1         1         1         1         1         1
  Modality 4 :

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    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000  0.0000  0.2699  0.6929  1.0000
Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 1.0000  1.0000  0.8748  1.0000  1.0000

--- Summary for K = 6 ---
Z.ar:
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.004444 0.015556 0.020000 0.021080 0.025000 0.085556

beta.ar:
Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.008333 0.035556 0.046442 0.070000 0.766667
Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1
Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1
Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1
Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1

gamma.ar:
Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.008889 0.039167 0.050373 0.077778 0.485000
Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.022222 0.630000 0.54810 1.000000 1.000000
Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.8339 1.0000  1.0000  1.0000  1.0000  1.0000
Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000  0.0000  0.2714  0.7210  1.0000
Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.8894  1.0000  0.8448  1.0000  1.0000

--- Summary for K = 7 ---
Z.ar:
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.009444 0.022778 0.029444 0.030751 0.036667 0.074444

beta.ar:
Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

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0.000000 0.005556 0.028333 0.043020 0.063333 0.756667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.031111	0.046923	0.071111	0.557222

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.3422	0.8189	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6206	1.0000	1.0000	0.9998	1.0000	1.0000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.1744491	0.0004167	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.5711	0.9283	0.7367	1.0000	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01778	0.03389	0.04056	0.04282	0.05000	0.12444

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.002222	0.015556	0.030921	0.047222	0.722778

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.002222 0.017778 0.034636 0.054444 0.491667
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000 0.0000 0.0000 0.2183 0.4806 1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.06611 1.00000 1.00000 0.99170 1.00000 1.00000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000 0.0000 0.0000 0.1027 0.0000 1.0000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000 0.0000 0.5156 0.4515 0.7861 1.0000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.3711 0.3933 0.4017 0.4024 0.4106 0.4728

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.003333 0.017778 0.036437 0.046111 0.762222
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
1 1 1 1 1 1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
1 1 1 1 1 1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
1 1 1 1 1 1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
1 1 1 1 1 1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.004444 0.020000 0.037400 0.052222 0.423889
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.00000 0.00000 0.00000 0.05161 0.00000 1.00000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000 0.4128 0.4661 0.4767 0.5228 1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.00000 0.00000 0.00000 0.02187 0.00000 0.95444
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.

```

```
0.0000 0.0000 0.1483 0.1915 0.3511 1.0000
', '\n`'))
```

```
## ' '
##
##
## --- Summary for K = 1 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.4089 0.4350 0.4428 0.4425 0.4511 0.4794
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.09111 0.13444 0.13070 0.16667 0.89667
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.09333 0.13778 0.13323 0.16944 0.79833
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.005000 0.008333 0.102017 0.031111 1.000000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0005556 0.0050000 0.0083333 0.0775255 0.0240278 1.0000000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.006111 0.015000 0.162485 0.118889 1.000000
##
## --- Summary for K = 2 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.1583 0.1839 0.1911 0.1914 0.1983 0.2228
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
```



```

## 0.00000 0.04778 0.09611 0.09797 0.13444 0.90111
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     1      1      1      1      1      1
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0500 0.1000 0.1006 0.1389 0.6800
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.001667 0.057778 0.414611 1.000000 1.000000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000000 0.0005556 0.0050000 0.2488552 0.3463889 1.0000000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.1233 1.0000 0.6974 1.0000 1.0000
##
## --- Summary for K = 3 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.04833 0.07111 0.07722 0.07806 0.08444 0.13111
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.03111 0.07389 0.07827 0.11167 0.90222
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     1      1      1      1      1      1
##
## gamma.ar:

```

```

## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.03333 0.07778 0.08166 0.11667 0.61222
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.5650  0.7033  0.6794  1.0000  1.0000
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.000000 0.002222 0.255317 0.431944 1.000000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.4156  1.0000  0.7528  1.0000  1.0000
##
## --- Summary for K = 4 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.01722 0.02944 0.03444 0.03493 0.03833 0.09722
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.02333 0.06111 0.06668 0.09444 0.76667
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.02500 0.06556 0.07076 0.10111 0.61278
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.7044  0.9494  0.7583  1.0000  1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.2793  0.8004  1.0000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  1.0000  1.0000  0.8672  1.0000  1.0000

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##
## --- Summary for K = 5 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.008889 0.024444 0.030000 0.031443 0.037222 0.105556
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.009444 0.038333 0.049916 0.076111 0.830556
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01000 0.04222 0.05349 0.08222 0.55333
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.3744  0.7767  0.6515  1.0000  1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.2699  0.6929  1.0000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  1.0000  1.0000  0.8748  1.0000  1.0000
##
## --- Summary for K = 6 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.004444 0.015556 0.020000 0.021080 0.025000 0.085556
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.008333 0.035556 0.046442 0.070000 0.766667
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

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##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.008889 0.039167 0.050373 0.077778 0.485000
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.022222 0.630000 0.54810 1.000000 1.000000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.8339 1.0000 1.0000 1.0000 1.0000 1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.0000 0.2714 0.7210 1.0000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.8894 1.0000 0.8448 1.0000 1.0000
##
## --- Summary for K = 7 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.009444 0.022778 0.029444 0.030751 0.036667 0.074444
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.005556 0.028333 0.043020 0.063333 0.756667
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.005556 0.031111 0.046923 0.071111 0.557222
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.3422 0.8189 1.0000

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## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.6206 1.0000  1.0000  0.9998  1.0000  1.0000
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000000 0.0000000 0.0000000 0.1744491 0.0004167 1.0000000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.5711  0.9283  0.7367  1.0000  1.0000
##
## --- Summary for K = 8 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.01778 0.03389 0.04056 0.04282 0.05000 0.12444
##
## beta.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.002222 0.015556 0.030921 0.047222 0.722778
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.002222 0.017778 0.034636 0.054444 0.491667
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.2183  0.4806  1.0000
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.06611 1.00000 1.00000 0.99170 1.00000 1.00000
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.1027  0.0000  1.0000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.5156  0.4515  0.7861  1.0000
##
## --- Summary for K = 9 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.3711  0.3933  0.4017  0.4024  0.4106  0.4728
##

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## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.000000 0.003333 0.017778 0.036437 0.046111 0.762222
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.000000 0.004444 0.020000 0.037400 0.052222 0.423889
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.05161 0.00000 1.00000
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.0000  0.4128  0.4661  0.4767  0.5228  1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.02187 0.00000 0.95444
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.1483  0.1915  0.3511  1.0000
##
## ‘ ‘ ‘

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Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1–9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.